Kinds as Projectibility Profiles

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Abstract

Anti-essentialist theories explain how categories can be real without essences. Homeostatic property cluster theory gives a Boydian answer, but HOMEOSTATIC now covers achievements that should be kept apart: stable projectible profiles, network order, maintenance, and corrective control. This paper gives anti-essentialist kind theory a stricter audit vocabulary. A category earns standing by tracking a projectibility profile: a worldly pattern whose partial observation licenses inferences under specified conditions. Profile analysis asks what the category projects, what supports the projection, what orders and stabilizes it, and where the claim should be demoted or rejected. Homeostatic control is a subcase, not the default.

Keywords: natural kinds; projectibility; homeostatic property clusters; kind claims; philosophy of science

1 Introduction

Anti-essentialist kind theory begins from a familiar pressure. Many categories that matter to science, grammar, social inquiry, and ordinary reasoning don't share an essence. Their members needn't have one property in common, and their boundaries may be graded, historically contingent, or context-sensitive. Still, some of these categories support reliable inference. They let us project from observed features to further expectations: to predict, explain, intervene, measure, compare, or learn.

Boyd's homeostatic property cluster framework offers one influential route through this pressure. A kind can be real when a cluster of properties supports induction and explanation because category use has been accommodated to worldly structure rather than because every member shares a defining essence (Boyd, 1991, 1999). That insight remains the starting point. But the word HOMEOSTATIC has been asked to do too much. In different contexts it can name a stable profile, a network of relations, a maintaining process, or a corrective-control system.

Those aren't the same achievement. A profile can be stable without a known maintainer. It can be ordered without being maintained. It can be maintained without being controlled. And it can support useful projection without earning the stricter label HOMEOSTATIC. Slater's stable property cluster account presses the stability point (Slater, 2015); Khalidi's causal-network account presses the

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order point (Khalidi, 2013, 2018); Weinberger's account of homeostasis and causal control presses the control point (Weinberger, 2026). The lesson isn't that Boyd was wrong, but that the evidential work needs sharper labels.

Call the label PROJECTIBILITY PROFILE. A profile, in this use, is the worldly property-and-relation pattern that makes partial observation a guide to further expectation within a stated range. The analysis of a profile specifies that pattern: which projection it supports, how we diagnose it, what makes the projected-from and projected-to properties co-available, what stabilizes that co-availability, and where the projection fails.

I use CATEGORY for the sorting item, PROFILE for the worldly pattern that supports projection, and KIND CLAIM for the assertion that the category earns standing by tracking such a profile. I use KIND inclusively for any projectibility profile. A NATURAL KIND claim and a HOMEOSTATIC claim are stronger, separable assertions about that support. The inclusive use doesn't make kindhood cheap. It shifts the burden from a single kind/non-kind threshold to what the category projects, under what scope, with what support, and with what demotion conditions.

Projectibility comes first methodologically and explanatorily. We start by asking what the category lets us infer, then ask what profile supports that inference, what relations make the profile non-accidental, what stabilizes it, and where it fails. That order doesn't imply that projectibility is metaphysically prior to every stabilizing structure. A mechanism earns its place in a kind claim by explaining why the declared projection holds, not by being named.[1]

The contribution is primarily regulative. This paper doesn't add a new entity to the ontology of kinds; it gives existing anti-essentialist resources an order of use: identify the projection, specify the profile, identify the ordering relation, state what stabilizes the profile, and declare scope and demotion conditions. It also keeps explanatory levels apart: a corpus distribution, a social practice, and a cognitive mechanism can't trade explanatory roles without an argued bridge.

Closest to this account is Ereshefsky and Reydon's grounded functionality account (Ereshefsky & Reydon, 2023). It ties a kind's standing to the aims it's posited for, and it blocks conventionalism with a world-side condition: a proposed category counts only when the relation it assumes is grounded in some aspect of the world, not merely in our conceptions of it. It also lets the function fix which aspect of the world must answer to it. On those structural questions, the accounts largely agree.

Where they part company is the licensed function. Inference here is broader than induction over shared properties.[2] Stable re-identification and historical individuation can still be projections: marker to re-identifiable group; membership to shared ancestry. With the function read that wide, fixing it as inference doesn't exclude Ereshefsky and Reydon's non-inductive cases; it redescribes them. The real contrast, then, is between sorting and projection: a class sorts, while a kind in good standing licenses projection. A grounded, aim-meeting category that licenses no further expectation is at most a thin class.

Two further gains follow. Their single grounding condition is split into an order axis and a stabilization axis: stabilizer, maintainer, and controller. The natural/non-natural boundary is replaced by support tiers and declared scope. A conventionally stabilized category such as CANADIAN PER-

[1] This paper doesn't take a stand on the stronger constitutive thesis sometimes associated with PROJECTIBILITY-FIRST formulations. Nothing in the argument depends on that stronger claim.

[2] Ereshefsky and Reydon (2023) use stable microbial re-identification and descent-based taxonomy against an induction requirement.

MANENT RESIDENT is a scoped kind here, projectible within its institution and demoted by scope overreach outside it. In that respect, the proposal is a constructive alternative to Ludwig (2018): keep kind talk, but replace the single kind/non-kind line with support tiers and scope conditions.

2 The Profile Package

A profile specification answers six questions. First, what's the projection target? Second, what pattern of properties and relations supports it? Third, what evidence diagnoses that pattern? Fourth, what ordering relation makes source and target properties co-available rather than accidental co-occurrences? Fifth, what stabilizes that co-availability across relevant contexts, timescales, or perturbations? Sixth, what scope conditions and failure modes mark the projection's limits?

Recurrence is cheap. A list of co-occurring properties becomes theoretically interesting only when it licenses some further expectation. The expectation may be predictive, explanatory, taxonomic, or intervention-guiding. It may concern a biological process, a grammatical pattern, a measurement practice, or a social role. But without a declared projection target, the category remains a description of observed association.

By PROFILE, I mean the property-and-relation pattern whose partial observation licenses the inference. The licensing runs in a direction: from the observed part of the profile to an expected feature, intervention, or decision. A symmetric association that supports no directed expectation isn't yet a projection in this sense.

In a biological case, a profile may include morphology, development, ecology, and reproductive relations. In a grammatical case, it may include distributional behaviour, semantic tendencies, diachronic sources, acquisition patterns, and mismatch residue. In a social case, it may include institutional marks, practical responses, uptake conditions, sanctions, and repair paths. The pattern needn't be one-level or one-mechanism.

Diagnostics come next, and they belong to the specification, not to the profile itself. They're the tests, traces, contrasts, measurements, or judgments by which inquiry locates the profile and checks whether a proposed projection travels beyond the cases that introduced it.

Non-accidental co-availability poses the structural question. Khalidi's causal network account supplies one important model: kinds aren't mere concatenations of properties, but nodes in ordered networks where some properties help explain others (Khalidi, 2018). In other domains, the relevant order may be functional, conventional, normative, or algorithmic rather than narrowly causal. Pluralism enters the explanatory description, not the evidential demand. Functional, conventional, and normative relations still have to be worldly dependences; the terms name forms that dependence can take, not weaker grades of support.

Here the issue is synchronic or structural: why are the projected-from and projected-to properties co-available in the cases at issue? In the broader use, the ordering relation still has to explain the declared projection; mere cue status isn't enough. Where the stronger Khalidian natural-kind claim is at stake, conventional or normative order needs causal, causal-historical, or causal-normative support.

The level has to be marked. Otherwise a social coordination pattern, a cognitive mechanism, and a corpus distribution begin to explain one another by equivocation.

Stabilization adds the modal or diachronic question. It asks what keeps the profile available across the relevant range of contexts, timescales, and perturbations. Some answers to the structural question also answer the stabilization question, but the claims differ. Network order says why co-availability isn't coincidence; stabilization says why it persists or recurs within the scope of the projection. Sometimes no separable stabilizer has been identified. Robust stability can still support limited projection, but stronger claims about maintenance or control require further support.

A STABILIZER explains why co-availability persists or recurs across the declared range. It may be a relation, constraint, mechanism, practice, or shared standard. Stabilizers needn't be active processes: selectional constraints, institutional practices, and ecological regularities can all stabilize a profile. What matters is that they explain why observing part of the profile gives reason to expect the rest.

A MAINTAINER is more specific. It's an active stabilizer that preserves or regenerates the profile over time or under perturbation. A maintainer may be developmental, social, institutional, biological, or interactional. It can keep a profile available without registering each departure as an error or returning a system to a fixed value.

A CONTROLLER is more specific again. It's a maintainer with perturbation-sensitive organization that preserves a higher-scale relation through lower-scale variation. On the stricter use defended in Reynolds (2026a), the term HOMEOSTATIC belongs here. Homeostasis isn't a synonym for recurrence, order, or maintenance. It marks the demanding case where corrective organization preserves or restores a relation despite perturbation. The restriction fits Weinberger's recent account of homeostasis and causal control, where feedback systems use lower-scale loops to produce higher-scale causal relations rather than simply holding variables constant (Weinberger, 2026).

One contrast keeps the levels apart. A valley's shape may be stabilized by erosion and gravity; wound closure is maintained by active healing; body temperature is controlled by corrective feedback. Only the last case is homeostatic in the strict sense.

Scope and failure complete the specification. A profile is projectible only across a declared range of conditions and contrasts, for a declared inferential purpose. The claim is field-relative in this limited sense: the inquiry selects which available projections matter, but it doesn't make them available. When the profile supports one family of inferences but not another, the result is a scoped claim: localize the projection, mark its defeats, and don't export it as a broader kind claim.

3 Why Projectibility Comes First

Goodman's new riddle made PROJECTIBILITY the name for a hard problem: some predicates that fit observed cases still don't license projection to unobserved ones (Goodman, 1955). This use of PROJECTIBLE is Boydian rather than Goodmanian in one central respect. It doesn't treat projectibility primarily as entrenchment in our predicate-using history. It asks what worldly order makes a category a reliable guide to further expectation.

This doesn't dissolve Goodman's problem. It presupposes a field-specific distinction between grounding order and case-fitting pattern, and it inherits local standards of explanation, dependence, and defeat. The aim is to regiment candidates, not to certify projectibility from outside the special sciences.

That realist turn is the point of Boyd's anti-essentialism. We don't need a shared essence to explain why a category supports induction. We need a non-accidental profile that our category use has partly learned to track (Boyd, 1991).

Projectibility is the target of the kind claim. Mechanisms, networks, histories, standards, and practices matter because they explain why projection works. This order blocks a common slide. Recurrence gets promoted to stability, stability to stabilization, stabilization to homeostasis, and homeostasis to kindhood. The profile vocabulary reverses that order. Start with the inference. Ask what profile licenses it. Then ask which stabilizing claims have actually been earned.

Starting with a mechanism makes the slide easier. Once a process has been named, the temptation is to let it certify the kind before asking what projection it improves. The profile-first order forces the mechanism to answer a narrower question: what inference would become weaker if that mechanism were removed?

Mechanisms may be primary in the world. During inquiry, though, we may have good evidence that a profile is projectible before we know what maintains it. The reliable projection still depends on whatever worldly relations make the profile non-accidental. Epistemic order and grounding order can come apart. The proposed discipline is to avoid treating the unknown ground as if it had already been identified, named, and promoted to homeostasis.

Stable property clusters fit the account for the same reason. Slater is right that natural kindness doesn't need to wait for a named homeostatic mechanism, and his stability demand is already projectibility-shaped: observing enough of a subcluster should license expectations about the larger cluster (Slater, 2015). Stability can be enough for a scoped kind claim when it licenses the declared projection. The profile question then asks which stronger claims about stabilizers, maintainers, or controllers have also been earned. A category may support modest local projections while failing to support broader ones. Both results are informative.

Purpose-relativity enters here with care. A profile is projectible for a family of questions, contrasts, and evidential practices. In physiology, a useful profile may license intervention and measurement. In grammar, it may license expectations about distributions, manipulations, and borderline cases. In social ontology, it may license expectations about uptake, status, sanctions, or institutional repair. The practice selects which projection matters. It doesn't make the projection true.

Availability is relative to operative stabilizers, which may be causal, conventional, or normative. Strong mind-independence isn't required: an institutionally stabilized profile can be available even when its stabilizers depend on us. What the account refuses is availability created by the inquirer's interest alone.

A projectibility profile differs from a checklist. Checklists record features. Profiles connect observed features to directed expectations under scope conditions. A checklist asks whether an item has properties P_1, P_2, and P_3. A profile asks whether observing P_1 and P_2, in the relevant setting, licenses expectation of P_3 rather than merely recording that all three co-occur. It also asks what would defeat that expectation.

That difference matters for explanatory economy. A mechanism-first paper can finish by naming structures that hold a cluster together. A profile-first paper has to say what those structures let us project, what failures would demote the claim, and whether the advertised stabilizer operates at the right explanatory level. The contribution is a discipline for using categories as guides to further expectation.

Here's a simple test for mechanism talk. If removing the mechanism leaves the same predictions in place, the mechanism hasn't earned its role in the kind claim. It may still be a true description of the system, but it's not yet doing the explanatory job a projectibility-profile analysis needs.

4 Profiles Across Levels

Profile vocabulary should travel without making every domain look biological. Profiles can be distributional, typological, model-internal, or social-practical, among others. Level matters because evidence at one level rarely licenses projection at another without a bridge claim.

Work on English definiteness supplies a concrete linguistic case. There, DEITALITY names a morphosyntactic profile for expressions that pattern with definite-marked noun phrases: *the* phrases, demonstratives, possessives, and some proper-name uses. The point isn't to rename every semantically definite expression. It's to ask whether those forms support further expectations: which diagnostics should cluster, where form and referent identifiability should come apart, which contrasts should shift under prosody or specificity manipulation, and which variation should appear across dialects and acquisition (Reynolds, 2025a). The category earns its place by those projections.

Typological comparison follows the same discipline. A comparative concept can remain an instrument for measurement. It licenses a naturalized kind claim only if the comparandum shows a profile that travels across languages, supports out-of-sample projection, and comes with demotion criteria when the profile proves too thin (Reynolds, 2025b). Here the stabilizing background may include acquisition, processing, diachrony, discourse pressure, and literate or institutional transmission. Measurement design supplies the audit trail, not a stabilizer. None of those should be collapsed into a universal language-internal category.

Truth-tracking in large language models supplies a different portability test. Rather than ask only whether self-consistency, retrieval, or tool use improves answers, the profile asks what an observed feature licenses us to expect next: citation accuracy, correction after retrieval, calibrated refusal, or successful use. A model may rely on heterogeneous supports, including output coherence, retrieved evidence, tool-mediated measurement, error correction, testimonial chains, and practical success (Reynolds, 2026b). Because those supports can come apart, a surface gain needn't be a truth-tracking gain.

In the document case, the projection is clearest. If answers grounded in documents fail by making unsupported claims, retrieval should improve citation accuracy where self-consistency improves only fluency. Self-consistency should help when the defect is incoherence among outputs. An evaluation profile would pre-register which support is being tested and what would count as failure. If agreement across samples rises while unsupported claims remain, the profile is coherence-stabilized, not truth-tracking, in that setting.

In each case, the same vocabulary applies without imposing the same mechanism. The DEITALITY profile is grammatical and distributional. The typological profile is comparative and methodological. The truth-tracking profile is epistemic and behavioural. Calling all three PROFILES doesn't erase those differences. It lets us ask the same disciplined questions: what gets projected, what pattern supports it, what stabilizes it, and where it fails.

Fields, practices, and institutions should be handled in the same way.[3]

They can be scope conditions or stabilizing backgrounds; in some social cases, institutional facts may also belong to the profile itself. Fields and practices shouldn't be promoted to kinds merely be-

[3] Boyd (1999, p. 148) already relativizes naturalness to a "disciplinary matrix", a family of inferential practices rather than an ordinary academic domain. This account keeps that relativity but separates Boyd's matrix from the social field or institution that carries the project. Boyd also says that accommodation demands are "themselves homeostatically related" (Boyd, 1999, p. 157); here that support claim has to be earned by evidence of corrective organization, not recurrence alone.

cause they organize inquiry or action. A field may condition which projections matter without itself being the projection target. A practice may stabilize the profile associated with a category without being the kind. Those are bridge claims, and they need their own evidence.

5 Failure Modes and Demotion

Vocabulary earns its keep only if it can say no. Without failure modes, projectibility profile would become another loose label for any interesting cluster. A profile view needs demotion rules: ways to say that a category is too thin, too local, wrongly levelled, or overpromoted.

The first failure mode is a thin profile. A thin profile has diagnostics that identify familiar cases but don't license further expectations. In grammatical work, the label negative polarity item can define a useful class: *ever*, *yet*, and *lift a finger* share sensitivity to negative contexts. The kind claim is thin, though, if the label doesn't predict the distinct licensing patterns of its members. In typology, a comparative label that can't generalize beyond the coding sample is thin.

The second failure mode is wrong-level projection. This happens when evidence at one level is used to project at another without a bridge. A corpus distribution doesn't by itself establish a model-internal mechanism. A social-practical regularity doesn't by itself establish a biological disposition. A grammatical category in one language doesn't by itself establish a cross-linguistic comparandum. The projection may be possible, but the bridge has to be argued.

The third failure mode is decorative mechanism. A paper names a mechanism, but the mechanism doesn't underwrite any new projection. The test from Section 3 returns here as a demotion rule: if removing the mechanism talk leaves the same predictions, the mechanism hasn't earned its place in the kind claim.

The fourth failure mode is over-homeostatization. Stability, network order, and maintenance are often real achievements. They shouldn't be relabelled as homeostasis unless the evidence shows corrective organization preserving a higher-scale relation through lower-scale variation. The risk is especially high outside biology, where the attraction of the homeostasis metaphor can hide weak evidence. A maintained convention, a stable distribution, or a reliable measurement practice may be projectible without being homeostatic.

The fifth failure mode is scope overreach. A profile can be real and useful under one set of conditions while failing elsewhere. The error is to export the projection without fresh tests. A category projectible for discourse analysis may not be projectible for syntax. A model behaviour projectible under benchmark conditions may not transfer to tool-using settings. A social status projectible inside one institution may not travel across institutions.

Demotion belongs inside realism. If kind claims are answerable to the projections they support, then failed projection should change the claim. Sometimes the right response is rejection: the category doesn't support the advertised inference. Sometimes localization is right: the category is projectible only under narrower scope conditions. Sometimes relevelling is right: the pattern is real, but at a different explanatory level.

Localization earns its keep only when the narrower scope is specifiable independently of the surviving cases: by domain, register, condition, or contrast fixed before the failures were seen. A scope drawn around exactly the successes redescribes failure as success; it doesn't demote the claim.

Profile talk joins promotion and demotion. A category earns stronger status when its profile supports reliable projection, when any claimed stabilizers are identified at the right level, and when its limits are known. It loses status when those conditions fail. The framework gives anti-essentialist kind theory an exit procedure, not just a route into realism.

6 Conclusion

More exactly, a kind claim earns standing when a category tracks a profile that licenses projection under specified conditions, with the ordering and stabilizing support the projection requires. The compact slogan is shorthand: category, profile, support, projection. Each element matters. No single essence need define the profile. Stabilization matters because recurrence needs explanation. Projectibility matters because kind claims earn their scientific or grammatical value by licensing further expectation.

Homeostasis remains important, but it no longer has to carry every achievement. Two axes have to stay apart. Network order is a structural claim about why co-availability isn't coincidence. Stabilization and maintenance concern persistence and recurrence; control adds corrective return after perturbation. A profile can be strong on one axis and weak on the other. Only the controller case deserves the strict homeostatic label. Keeping those distinctions apart preserves Boyd's anti-essentialist realism while incorporating Slater's stability-first lesson and Khalidi's network discipline.

Several grounding questions remain open. The framework doesn't decide in advance whether a projection is grounded causally, conventionally, normatively, or by some mixture of supports. It doesn't solve Goodman's new riddle. And it doesn't assume that all six profile questions will receive strong answers in every case. It gives the order in which those answers have to be earned.

So the vocabulary leaves five questions. What does this category let us infer? What pattern supports that inference? What stabilizes the pattern? What would defeat the projection? And when should the category be demoted, localized, or rejected?

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